

PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 15, 2025
Status: [REDACTED]
Tracking No. m8b-21br-ld4x
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001

Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479

Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-7952

Comment on FR Doc # 2025-02305

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General Comment

I am a software engineer specializing in high-performance computing. I have two pieces of input.

First, machine learning models must not be permitted to train on unlicensed content. Recent research such as Carlini et al.'s "Extracting Training Data from Diffusion Models" [2023, <https://arxiv.org/abs/2301.13188>] demonstrates that diffusion models (the current state of the art for image generation) can memorize and regurgitate their training data. Nasr et al.'s "Extracting Training Data from ChatGPT" [2023, <https://arxiv.org/abs/2311.17035>] shows the same for text-based Large Language Models. Systems that produce substantial copies, that substitute for the original works, and which are operated commercially, may well fail the principles of fair use, harm American creators, and should not be exempted from copyright requirements.

Secondly, machine learning models should apply robust, visible and machine-readable watermarks to their output. Shumailov et al. [2023, <https://arxiv.org/abs/2305.17493>] demonstrated that LLM quality degrades as a higher fraction of their input is produced by other LLM models. As an increasing amount of the text on the Internet is generated by LLMs and several ML models are trained indiscriminately on Internet content with little or no way to filter out LLM content, we may well see a "model collapse" situation: models of 2026 and later may be lower-quality than those of 2025, because the input corpus has become so contaminated. Similarly, robust watermarking would help guard against impersonation attacks, like those of "deepfake" models. Models such as these allow impersonating of individuals' appearances and voices, damaging the credibility of public figures.

It's important for me to mention that the above statements and opinions are my own, and may not represent those of my employer.

Thank you.